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1.

**1.1 Context of Use**

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Context

Where users could be:

Farmers: On their farms or at their homes in rural Eastern Cape, often in areas with poor road infrastructure, limited internet connectivity and basic smartphones.

Traders (hawkers, market vendors and spaza shop owners): At their stalls, shops or markets in nearby towns or urban areas, often managing customers while searching for fresh produce.

Transport Providers: Travelling between rural farms and urban areas, or waiting at collection points, where they need quick access to transport requests.

Vulnerable Households: At home or within their local communities, using the platform to look for available food or request assistance when needed.

What users could be doing while using the solution:

Farmers: Using the platform between farming activities to list surplus produce, update availability or respond to requests.

Traders: Searching for available produce while serving customers or planning stock for their businesses.

Transport Providers: Checking available transport requests while planning delivery routes or waiting for collection jobs. ✓

Vulnerable Households: Browsing available food, requesting produce or checking notifications about collections or deliveries, especially during times of food insecurity.

**1.2 Expected Usability Goals**

Usability

Effectiveness – The platform should help rural farmers successfully connect with traders, transport providers and households so that surplus produce reaches people who need it instead of being wasted.

Efficiency – The platform should allow users to complete tasks such as listing produce, searching for available food and accepting transport requests with as little time, effort and travel as possible.

Utility – The system should provide the essential functions needed for food distribution, including listing produce, searching for available food, requesting transport and confirming deliveries.

Learnability - The platform should be simple and easy to use for people with limited digital literacy by using clear instructions and straightforward navigation.

Safety: The solution should reduce the risk of wasted trips, spoiled produce and failed collections by providing accurate information about produce availability, collection locations and delivery arrangements.

Memorability: The interface should be consistent and easy to remember so that users who only access the system occasionally can use it without relearning how it works. ✓

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**Expected User Experience (UX)**

UX

-The proposed solution should provide a positive and satisfying user experience for all user groups. Farmers should feel empowered and hopeful because they can reduce food waste and increase their income opportunities. Vulnerable households should experience reassurance and reduced anxiety by having better access to available food.

- The platform should create a trustworthy, convenient and supportive experience that encourages continued use and strengthens collaboration between all stakeholders.

**1.3 Tasks users need to complete**

Tasks

-Farmers: Create an account and log in; list surplus produce by entering the crop type, quantity, location and whether it is for sale or donation; update produce availability; remove or close listings once the produce has been collected.

Traders (hawkers, market vendors and spaza shop owners): Browse or search for available produce from nearby farmers; view produce details; request or reserve produce; arrange collection or delivery; confirm receipt of the produce.

Transport Providers: View available transport requests, including collection and delivery locations and the type and quantity of produce; accept or decline transport jobs; update delivery status; confirm when the delivery has been completed.

Vulnerable Households: Search or browse available food nearby; request or reserve available produce; receive notifications about collection or delivery; confirm receipt of the food.

**1.4 Pre-existing Knowledge Users Should Have**

all users? −2

Knowledge

- Users should have basic knowledge of operating a smartphone or computer, including opening an application, navigating menus and entering information into simple forms. They should understand how to create an account, log in and read notifications. Farmers should know basic information about the produce they wish to advertise, such as the crop type, quantity and availability. No advanced technical knowledge should be required because the system is intended for users with varying levels of digital literacy.

Not technical know-how

**1.5 Two Relevant Interface Metaphors**

Metaphors

Marketplace - The application can be organised like a traditional market where farmers display their produce, buyers browse available products and transactions take place between sellers and buyers. This familiar concept makes the system easier to understand.

Shopping basket/Cart - Users can add the produce they want to collect or request into a basket before confirming their request. This metaphor is widely used in online shopping and helps users understand how to manage their selected items.

**1.6 Two Relevant Interaction Types**

Interaction

how? What makes this instructing? What does the user do?

Instructing - Users perform actions by giving commands to the system, such as adding produce, searching for products, requesting deliveries or updating information. The system responds by carrying out the requested action.

No – not conversing with other users. Should be with the system.

Conversing - The system allows users to communicate with one another through messages or chat to arrange collections, deliveries or answer questions about available produce. The system

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may also provide automated responses or notifications, creating a two-way interaction that supports communication between users.

2. a)

[13]

- Deepseek
- Suggest a digital solution that helps reduce food waste by connecting small-scale farmers with people, businesses, and organisations that need fresh produce. Describe the main tasks users can perform and explain how the solution addresses usability goals.
- To address food waste and hunger in the Eastern Cape, a mobile platform can connect small-scale farmers with surplus produce to buyers like spaza shops, hawkers, and clinics, while also incorporating transport providers and community organizations. Farmers could list surplus, accept orders, and request transport, while buyers search for produce and place orders, with transport providers managing deliveries. Usability would be enhanced through a bilingual interface, voice-enabled listing, and WhatsApp support, while real-time matching reduces spoilage and costs. An escrow system ensures fair payments, and a donation channel with tax incentives encourages giving instead of wasting, easing farmers' frustration. Offline functionality and SMS backup address connectivity issues, while user ratings and verification build trust. This practical solution would reduce waste, generate income, and feed hungry families across the province.

b)

- Most of the tasks proposed by DeepSeek are relevant to the assignment scenario. These include listing surplus produce, placing orders, arranging transport, and matching farmers with buyers, as they help improve food distribution and reduce waste. The tool also suggested tasks that we did not identify, such as an escrow system for secure payments, a donation channel with tax incentives, and a user rating system. We believe these are relevant because they promote trust, fairness, and encourage food donations. However, the tool did not include the task we identified of conducting research to determine whether a digital platform is accessible and suitable for users, considering network availability, smartphone access, data costs, and digital literacy. This is important to ensure the solution is practical for rural communities. ✓

  That was not what you were asked to check for. It is about tasks to be performed by the user using the proposed solution —2
- The usability goals identified by DeepSeek map well to the usability goals we studied. Features such as voice support, a bilingual interface, and offline functionality improve learnability and effectiveness. Real-time matching improves efficiency, while secure payments and user verification support safety. The platform also provides the necessary functions for advertising, ordering, and transporting produce, meeting the utility goal. Although memorability was not explicitly mentioned, the platform's simple and consistent design would make it easier for users to remember how to use it. ✓

3) **a) <u>Memory</u>**

[10]

**From Assignment 1:** "A potential solution is a central digital platform that enables farmers to find buyers for their surplus produce, traders and households to find available food, and transport providers to coordinate the collection and delivery of produce."

Memory is important because users should not have to remember complex procedures when using the platform. The interface should use familiar icons, clear labels, consistent menus, and simple navigation to reduce cognitive load. It should also promote recognition rather than recall, making the platform easier to use for users with different levels of digital literacy.

**Attention**

**From Assignment 1:** "Research would be conducted to determine whether the platform is accessible and appropriate for users, considering network availability, smartphone access, data costs, and digital literacy."

Attention is important because users need to quickly identify relevant information, such as available produce, transport requests, and order updates. The platform should use headings, colour, spacing, and clear notifications to highlight important information while avoiding information overload. This helps users stay focused and complete tasks efficiently.

- Memory and attention are closely related. Users must first pay attention to information before it can be stored in memory. As users become familiar with the platform, memory helps them recognise features and complete tasks more quickly, while attention helps them focus on important updates and tasks. Together, these cognitive processes improve usability and provide a better user experience.

4)

a)

**Game: Fortnite**

(8)

Fortnite is a multiplayer online battle royale game in which up to 100 players compete against one another. Players can play individually or in teams, with the main objective is to be the last one standing on the battle royal island! Loot, build, explore, and fight to overcome other players in the match

Is this your description of the goal?

Several social protocols and conventions are used in Fortnite. Players communicate using voice chat or text chat to coordinate strategies. Team members take turns speaking so that everyone can share information, which is like conversational turn taking. Players collaborate by sharing weapons, ammunition and healing items. They can revive each other when someone is knocked down. Players maintain awareness of teammates by monitoring their locations, health and actions on the game, demonstrating co presence. For nonverbal communication players use pings, markers and emotes

b)

Players interact using game controllers, keyboards or touch controls to move, build structures, collect resources and fight opponents. Communication occurs through voice chat, text chat, map markers (pings), emotes and visual indicators. These communication methods allow players to coordinate strategies, warn teammates about danger and share information in real time. This supports collaboration and social interaction among players

One social phenomenon is that players often communicate more openly with strangers from different countries because they are interacting online rather than face-to-face. Another phenomenon is the formation of temporary online teams and communities that may only exist during gameplay. Players may also experience peer pressure, competition and collaboration within virtual communities, which are common in online social games. These behaviours occur because technology enables people to interact remotely, creating a sense of social presence and community even though they are physically apart